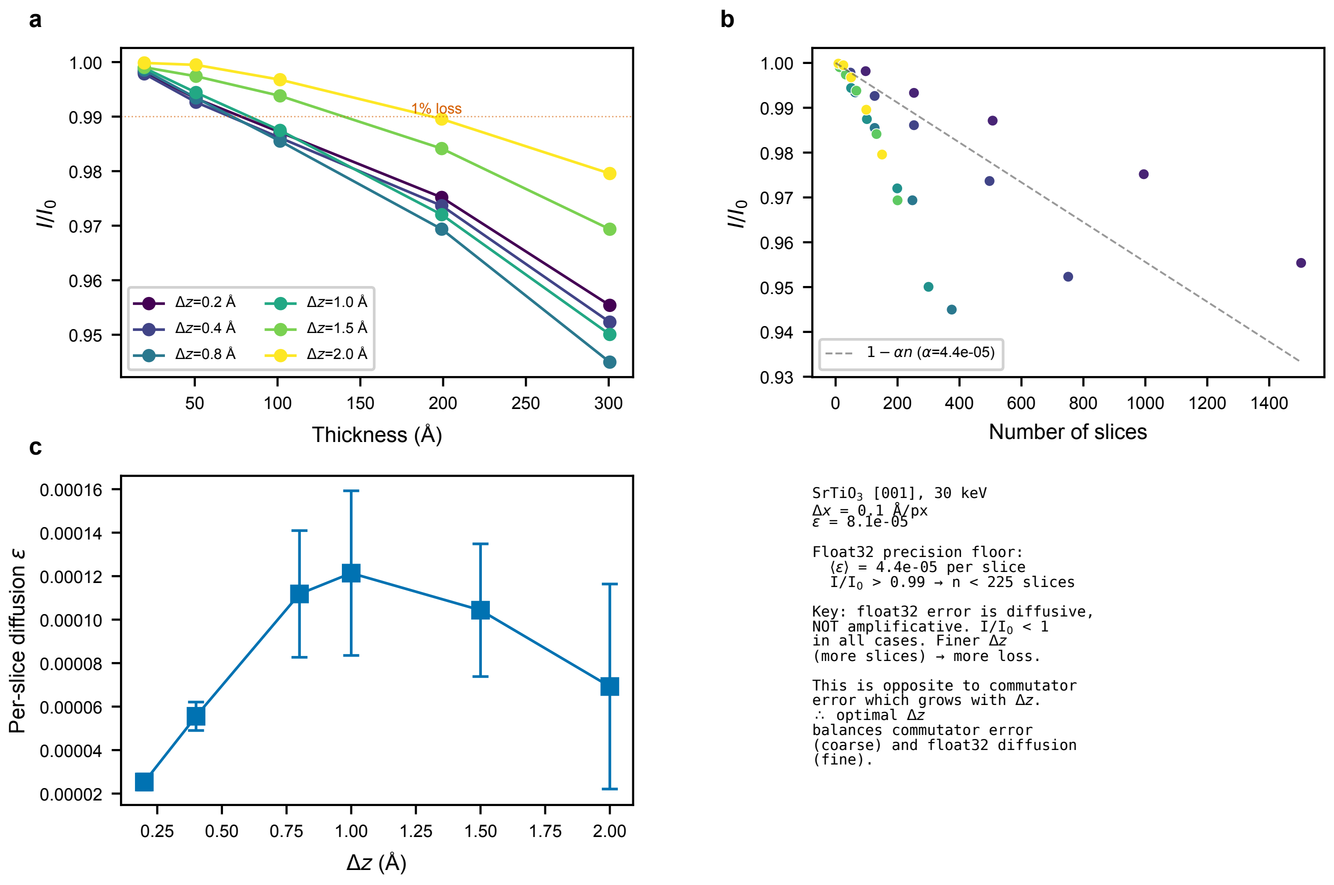
a
1.00
0.99
0.98
0.97
0.96
0.95
I/I0
1% loss
Δz=0.2 Å
Δz=0.4 Å
Δz=0.8 Å
Δz=1.0 Å
Δz=1.5 Å
Δz=2.0 Å
50
100
150
200
250
300
Thickness (Å)
b
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
I/I0
1 − αn (α=4.4e-05)
0
200
400
600
800
1000
1200
1400
Number of slices
c
0.00016
0.00014
0.00012
0.00010
0.00008
0.00006
0.00004
0.00002
Per-slice diffusion ε
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Δz (Å)
SrTiO3 [001], 30 keV
Δx = 0.1 Å/px
ε = 8.1e-05
Float32 precision floor:
⟨ε⟩ = 4.4e-05 per slice
I/I0 > 0.99 → n < 225 slices
Key: float32 error is diffusive,
NOT amplificative. I/I0 < 1
in all cases. Finer Δz
(more slices) → more loss.
This is opposite to commutator
error which grows with Δz.
∴ optimal Δz
balances commutator error
(coarse) and float32 diffusion
(fine).